

6CCS3PRJ

Designing for Consent Revocation in Online Platforms

Final Project Report

Author: Mert Ayranci

Supervisor: Kovila Coopamootoo

Student ID: 21175534

March 15, 2026

Abstract

The abstract is a very brief summary of the report's contents. It should be about half-a-page long. Somebody unfamiliar with your project should have a good idea of what your work is about by reading the abstract alone.

Originality Avowal

I verify that I am the sole author of this report, except where explicitly stated to the contrary. I grant the right to King's College London to make paper and electronic copies of the submitted work for purposes of marking, plagiarism detection and archival, and to upload a copy of the work to Turnitin or another trusted plagiarism detection service. I confirm this report does not exceed 25,000 words.

Mert Ayranci

March 15, 2026

Acknowledgements

It is usual to thank those individuals who have provided particularly useful assistance, technical or otherwise, during
your project. Your supervisor will obviously be pleased to be acknowledged as he or she will have invested quite a lot of time overseeing your progress.

Contents

1 Introduction **3**

1.1 Motivations 4
1.2 Scope 4
1.3 Research Questions 5
1.4 Aim and Objectives 5
1.5 Contributions 6
1.6 Key Takeaways 7
1.7 Report Structure 7

2 Literature Review **9**

2.1 Brief 9
2.2 Digital Messaging and Photo Sharing 9
2.3 Digital Privacy in Modern Communication 10
2.4 The Concept of Consent 13
2.5 Consent in Instant Messaging and Photo Sharing 16
2.6 Human Computer Interaction (HCI) and Interface Design 20
2.7 Empirical Usability Assessments of Current Platforms 23
2.8 Research Gap and Summary 24

3 Requirements **27**

3.1 Brief 27
3.2 Stakeholders 27
3.3 Prototype Scope (tekrar bakilacak) 28
3.4 Use Cases 28
3.5 Functional Requirements (FR) 31
3.6 Non-Functional Requirements (NFR) 39
3.7 Feasible Extensions Consistent with Current Scope 45

4 Design & Specification **47**

4.1 Brief 47
4.2 Design Rationale and Principles 47
4.3 Conceptual System Model: Smart Photos 51
4.4 Functional Specification 52
4.5 Non-Functional Specification 57

5 Implementation **61**
5.1 Section Heading 61

6 Legal, Social, Ethical and Professional Issues **62**
6.1 Section Heading 62

7 Results/Evaluation **63**
7.1 Software Testing 63
7.2 Section Heading 63

8 Conclusion and Future Work **64**
Bibliography 70

A Extra Information **71**
A.1 Tables, proofs, graphs, test cases, ... 71
A.2 Survey Information Sheet 71
A.3 Survey Questions 71

B User Guide **72**
B.1 Instructions 72

C Source Code **73**
C.1 Instructions 73

Chapter 1

Introduction

The widespread use of digital messaging platforms has transformed how people share personal images, maintain relationships, and negotiate boundaries in everyday communication. Despite developments in security safeguards, users continue to face substantial difficulties in controlling shared content once it reaches another person's device. Current messaging platforms provide protection against external threats but offer limited tools for managing interpersonal risks such as regret, changing relationships, or misuse of personal photos (Trepte, 2020). These vulnerabilities are especially concerning given that consent in digital contexts is frequently dynamic rather than static, and users may seek to modify or withdraw access to shared material over time. Major messaging applications such as WhatsApp, with over 3 billion monthly active users (Statista, 2025), employ end-to-end encryption to secure message content. Yet this model does not extend to post-delivery control between users, leaving senders without meaningful mechanisms to revoke or modify consent once a photo has been delivered. While recent studies in privacy, digital safety, and interpersonal communication emphasise the significance of modifiable consent, practical design solutions for major messaging apps remain underdeveloped. Further work is needed to explore interaction models that support revocable consent without disrupting existing communication patterns. Although WhatsApp is used as the primary case study, the user needs and interaction principles explored in this project are intended to generalise to other messaging platforms that support private photo sharing and forwarding.

1.1 Motivations

Personal photos have become a common method of expression and interaction in digital communication, with users sharing them on a regular basis across WhatsApp. However, once an image is sent, users often lose the ability to influence how it will be used, stored, or redistributed. This lack of post-sharing control creates a disconnect between the technical protections offered by WhatsApp and the changing, context-dependent nature of interpersonal consent. While end-to-end encryption protects messages from external access, it does not address how users manage privacy between one another, nor does it accommodate situations where consent evolves over time.

There exists researches to some extent across digital privacy, human–computer interaction, and online safety, highlighting increasing concerns about image misuse and the loss of control users experience after sharing personal content. However, the practical design of mechanisms to prevent these issues, particularly in the context of dynamic consent, remains underexplored. Most existing work on consent revocation focuses on legal or conceptual frameworks rather than on usable, interface-level solutions within everyday messaging platforms such as WhatsApp. This project is motivated by the need to address this gap by examining how WhatsApp can better support ongoing, revocable consent when users share photos.

1.2 Scope

This project examines post-sharing consent revocation for personal photos in WhatsApp across multiple communication contexts, including one-to-one private chats, group messaging, and semi-public spaces such as communities. The primary focus is on how users can modify or withdraw consent after sharing photos, and how interface design can support this across different audience sizes and social dynamics. The scope is limited to image-based content; calls, video calls, videos, voice notes, documents, and status updates are excluded. The research is confined to user experience and interface design, investigating user-facing interaction patterns rather than underlying system architecture or encryption mechanisms. Large-scale user testing, behavioural studies, and technical feasibility assessments fall outside the project scope.

1.3 Research Questions

This project is guided by a set of research questions that aim to investigate how consent revocation can be effectively supported within WhatsApp's messaging environment. These questions address the limitations of current functionality, the expectations and concerns of users, and the design requirements for implementing revocable consent in a way that is understandable, usable, and trustworthy. Together, they shape the direction of the background research, design specification, prototype development, and evaluation.

The research questions are as follows (1 main question, 5 supporting questions):

Main research question: How can WhatsApp be designed to support clear, usable, and trustworthy mechanisms for revoking or modifying consent after users share photos in private conversations?

1. What limitations exist in WhatsApp's current design for managing photo sharing, and how do these limitations affect users' ability to control shared content?

2. What expectations, concerns, and behaviours do WhatsApp users have regarding post-sharing control of personal photos?

3. Which interface elements and interaction patterns best communicate photo permissions and make revocation options understandable within WhatsApp's existing UI?

4. How can the process of revoking or modifying consent be designed to be intuitive and fair for both the sender and the recipient?

5. How do the proposed revocation features affect the' perceived trust, safety, and sense of control of users when sharing photos on WhatsApp?

1.4 Aim and Objectives

The aim of this project is to develop and assess user interface mechanisms that facilitate revocable, ongoing consent within WhatsApp's photo-sharing environment, allowing users to maintain control over shared content in a transparent, intuitive, and reliable manner.

To achieve this aim, the project pursues the following objectives:

1. Analyze the limitations of the current WhatsApp design in relation to the control of photo-sharing.

2. Investigate user expectations, concerns, and behaviours with respect to post-sharing consent.

3. Define functional and non-functional requirements for effective consent revocation features.

4. Design a set of interface components and interaction flows that demonstrate how revocable consent can be integrated into the WhatsApp user interface.

5. Develop a high-fidelity prototype that illustrates the full permission and revocation workflow.

6. Evaluate the usability, clarity, trustworthiness, and perceived fairness of the proposed features through user testing.

7. Reflect on the broader implications for user safety, privacy, and ethical communication design.

1.5 Contributions

The project makes the following contributions:

1. A high-fidelity interactive Figma prototype demonstrating a redesigned consent revocation system for photo sharing in WhatsApp-like messaging environments.

2. A set of realistic interaction flows, UI states, and permission management behaviours modelling how revocable consent could function in practice.

3. A usability evaluation of the proposed design, assessing clarity, perceived trustworthiness, and fairness for both senders and recipients.

4. A human-centred design analysis of the practical implications of enabling revocable consent for photo sharing in messaging platforms.

1.6 Key Takeaways

1.7 Report Structure

!TEX root = ../report.tex

Chapter 2

Literature Review

2.1 Brief

This chapter examines the literature on how the design of online messaging platforms shapes users' ability to understand, manage, and revoke consent, with a specific focus on photo sharing on WhatsApp. It begins with a brief overview of digital messaging and photo-based communication to situate the social and privacy-sensitive nature of image sharing. It then reviews relevant theoretical and consent-related perspectives to explain why consent revocation is not only a technical issue but also a contextual and interpersonal one. The chapter subsequently examines research on instant messaging, photo sharing, and interface design to assess how current platform features support or constrain meaningful user control. Finally, it synthesises the literature to identify research gaps that motivate the proposed user-centred revocation design.

2.2 Digital Messaging and Photo Sharing

Digital messaging has developed from earlier forms of computer-mediated communication into mobile, app-based, and increasingly multimodal communication environments, where text, images, and other media are integrated within the same platforms. Herring's historical overview highlights this shift from predominantly text-based CMC toward convergent and multimodal communication systems, providing a useful background for understanding contemporary messaging platforms [16]. Research on mobile messaging also shows how texting became embedded in everyday social contact, particularly in peer communication shaped by social norms and

situational constraints [20]. In parallel, early camera-phone studies show that images were used not only for personal capture but also for communication with absent others and remote tasks, helping establish photo messaging as a routine social practice [23]. This transition is important because it created communication settings in which image sharing carries social meaning and privacy expectations that may later complicate consent revocation.

2.3 Digital Privacy in Modern Communication

This section grounds the dissertation in theories that explain why consent revocation is not simply a delete button, but a socially negotiated and context-dependent action.

2.3.1 Communication Privacy Management (CPM)

Communication Privacy Management (CPM) theory explains privacy as an ongoing process of boundary management rather than a fixed state, making it useful for analysing disclosure in interpersonal and mediated communication [9, 31]. Petronio and Child (2020) describe CPM as a framework in which individuals treat private information as something they own and control, and they manage access to it through privacy rules shaped by context, relationship, perceived risk, and expectations about appropriate use after disclosure [31]. A key concept is co-ownership: once information is shared, recipients may become legitimate co-owners who develop their own assumptions about how that information can be retained, discussed, or circulated [31]. CPM therefore shifts attention from information itself to the negotiated rules and expectations that organise its flow.

This matters for consent revocation in messaging because consent revocation in messaging is not only a technical question of whether content can be deleted; it is also a social question about whether previously established privacy expectations can be changed after disclosure [9, 31]. CPM helps explain why post-sharing control is often contentious: a sender may understand revocation as a reassertion of personal control, while a recipient may interpret the same act as changing the rules after access has already been granted [31]. Hence, CPM provides a strong theoretical foundation for studying consent and control in messaging-based photo sharing, where interpersonal expectations are central to how users judge fairness, trust, and appropriateness.

Applied to revocable or dynamic consent in messaging, CPM suggests that revocation should be understood as an attempt to re-coordinate privacy boundaries rather than simply erasing data

[31]. The theory's concept of boundary turbulence is especially relevant here, as turbulence occurs when co-owners hold conflicting privacy rules or when expected information flows break down [31]. In messaging environments such as WhatsApp-style photo sharing, this framework helps explain why consent may need to be updated over time and why platform mechanisms for revocation must operate within the tension between social boundary expectations and the technical ease of retention, copying, or redistribution [9, 31].

2.3.2 Contextual Integrity (CI)

Contextual Integrity (CI) reframes privacy as appropriateness of information flows, not simply secrecy or access control [29]. Nissenbaum (2011) argues that information sharing is governed by contextual norms: who is involved (actors), what is shared (attributes), and how it moves (transmission principles) [29]. This is directly relevant to WhatsApp photo sharing because users ofMNten treat a private chat as a bounded context with an implied "right audience," even when the content is not technically protected from recipient-side redistribution. Under CI, harm occurs when a photo's flow changes context (e.g., from a one-to-one chat into a group, another platform, or offline circulation) because the transfer violates the original contextual expectations, even if the sender originally shared willingly [29].

CI is also useful for critiquing the limits of the notice-and-consent paradigm in everyday messaging. Nissenbaum (2011) critiques notice-and-consent as insufficient for online privacy governance, while Bourgeus and Vandercruysse (2024) argue that overreliance on consent often fails to produce meaningful privacy control in complex data ecosystems and motivate contextual and communal alternatives [5, 29]. This matters for photo sharing because images frequently involve group privacy: a photo can affect the sender, the recipient, and any depicted co-subjects, each with potentially different expectations about appropriate sharing [5]. Design implication: revocation mechanisms should be framed as restoring context-appropriate flow (limiting further redistribution and future access within WhatsApp), and the UI should represent that permissions apply to a social context (audience and transmission rules), not just to an individual recipient or a one-time send action.

2.3.3 Media Richness Theory (MRT) and Social Presence

Media Richness Theory (MRT) holds that communication media vary in their capacity to convey information in ways that support interpretation and mutual understanding, and that

richer media are generally better suited to tasks involving ambiguity (equivocality) [10]. In Daft and Lengel's account, rich media allow more immediate feedback and convey a greater variety of cues, which can help communicators clarify meaning and reduce misunderstanding when issues are socially sensitive or open to multiple interpretations [10]. WhatsApp-style messaging that includes photo sharing can therefore be treated as relatively rich in everyday practice because it combines rapid interaction with visual content that can communicate relational and emotional information. At the same time, empirical work has qualified strong deterministic readings of MRT: Dennis and Kinney (1998) report that while cue and feedback properties shape perceptions of media richness, matching media richness to task equivocality did not reliably improve performance outcomes in their study [11].

Social Presence Theory provides a complementary lens by emphasising the interpersonal qualities of mediated communication. Short, Williams, and Christie (1976) conceptualise social presence as the extent to which a medium conveys the salience of the interaction partner and the sense of interpersonal involvement, with higher social presence generally associated with greater warmth and immediacy in interaction [37]. In contemporary messaging contexts, this helps explain why users may prefer channels that support richer interpersonal signalling for relational communication, even when such channels also create additional privacy and control demands. Ishii et al. (2019) further argue that media richness should be revisited for modern technologies because users' experiences of richness are shaped not only by channel properties but also by evolving features and patterns of use, which is particularly relevant for mobile messaging environments [19].

Taken together, MRT and Social Presence Theory help explain why revocable or dynamic consent in messaging is not only a technical problem but also a socially consequential one. When users select richer media for intimate or socially sensitive exchange, the interaction may carry stronger interpersonal involvement and relational expectations about how shared content ought to be handled [10, 37]. In such contexts, withdrawing consent may be experienced as a delicate change in interpersonal boundaries rather than as a neutral technical action, because it can signal mistrust or create a sense of relational distance. Design implication: revocation mechanisms in WhatsApp-style photo sharing should be designed with attention to this social meaning, supporting face-saving and relationship maintenance while clearly communicating the practical limits of technical control in the presence of retention and recirculation practices [11, 19].

2.4 The Concept of Consent

This section examines consent in digital communication and argues that, in messaging contexts, it should be understood as an ongoing process rather than a one-time decision. While legal frameworks define the formal conditions under which consent is considered valid, consent in interpersonal messaging also functions as a relational and contextual expectation about what others may appropriately do with shared content.

2.4.1 The Legal Mandate (Bunu intro motivation kismina alabilirsin??)

Although the GDPR is not designed specifically for consent revocation in online messaging platforms, it provides a useful benchmark for how valid consent and withdrawal should be implemented in such environments. Under the GDPR, consent must be freely given, specific, informed, and unambiguous, and withdrawal must be as easy as giving consent; where consent is withdrawn and no other legal basis remains, the right to erasure may also be engaged [12, 32]. Chaurasiya (2025) shows that the Indian DPDP Act adopts a similar consent-centred approach by defining consent in affirmative terms and recognising mechanisms for managing and withdrawing it [7]. However, as Politou et al. (2018) note, revocation operates prospectively rather than retroactively, and erasure becomes technically difficult once data have been copied, shared, or replicated across systems [32]. In distributed communication contexts such as messaging, the legal mandate therefore establishes a right to revoke consent, but not a guarantee of complete practical forgetting.

2.4.2 Consent Fatigue, the Validity Gap, and Non-Use

Although legal and policy frameworks often treat consent as meaningful once it has been formally obtained, this assumption is difficult to sustain in contemporary digital environments. Baruh and Popescu argue that privacy self-management based on notice and choice places too much responsibility on individuals, even though data practices are often opaque, complex, and difficult to resist in practice [3]. In such settings, consent may be legally valid without being substantively meaningful, creating a validity gap between formal agreement and real user control. They further show that users may respond in two limited ways: through partial withdrawal or avoidance when privacy risks feel unmanageable, or through assimilation, where users accept reduced control and rely on market-provided protections [3].

Waldman extends this critique by showing that privacy is often operationalised narrowly within

organisations. In his account, lawyers tend to reduce privacy to notice and choice, while engineers often equate it with security or encryption, leaving limited space for more user-centred understandings of consent and control [39]. For the present topic, this suggests that effective revocation cannot depend only on formal consent or interface disclosure, but must be supported by designs that make control realistic and usable in practice.

2.4.3 Consent Models

Consent in everyday messaging cannot be understood adequately as a one-off, individualised act of permission, because interpersonal communication unfolds over time, across relationships, and under conditions of partial and often irreversible loss of control. Most of the revised consent models discussed below were developed in health, research, or broader data-governance settings rather than instant messaging; they are used here analytically, as conceptual tools for thinking about permission, revision, and withdrawal in WhatsApp-style communication. Traditional consent frameworks tend to assume that consent is static, singular, and transaction-like, yet this logic becomes unstable in ongoing interpersonal exchange, where the meaning of permission can shift after disclosure. As Burkhardt et al. suggest, meaningful consent depends not only on formal agreement but also on understanding and perceived control, while work on consent and revocation has also shown that decisions made at one moment cannot remain fully informed across the lifetime of data use [1, 6]. This is why dynamic consent has been positioned as a response to one-off authorisation models that ask people to agree to future uses they may neither anticipate nor revisit [36]. In messaging, these weaknesses are particularly visible because permission is shaped by repeated interaction, changing context, recipient interpretation, and what happens after content has already been sent.

Dynamic Consent offers the clearest correction to this temporal problem by treating permission as revisable rather than fixed. Across the literature, it is characterised by ongoing engagement, granular choice, and the possibility of adjusting preferences over time rather than accepting a single all-or-nothing decision [22, 24]. This broader orientation is reinforced by Wilbanks' argument that digital consent should be designed as the beginning of an ongoing ethical relationship rather than a one-time administrative step, and by later work that frames modifiable consent choices as central to user-centred data governance [28, 47]. However, what Dynamic Consent improves most clearly is temporality, not post-disclosure control. In everyday messaging, the ability to revise a preference does not necessarily translate into the ability to affect already-delivered content once it has moved beyond the sender's immediate control.

Revocable Consent is therefore closer to the central concern of this dissertation, because it foregrounds withdrawal after permission has already been granted. The literature usefully distinguishes between revocability as a normative requirement and revocation as a practically enforceable outcome. Consent and revocation have been framed as ongoing controls, with informed revocation emphasising that users should understand the forms of withdrawal available to them [1]. Related technical work similarly treats withdrawal as a change in the scope of authorised access: users should be able to specify who may access data, for what purposes, and for how long, while formal models of consent evolution represent revocation by closing a previously open interval of collection or access rather than undoing prior disclosure [27, 33]. This distinction is especially important for photo sharing, where a sender may be able to end future authorised access without being able to retract a photo that has already been viewed, saved, or copied by a recipient. Revocation in messaging is therefore best understood as limiting future legitimacy and platform-supported access, not as guaranteeing recovery of the original disclosure. This problem is reinforced by the broader notice-and-consent literature, which shows that even ostensibly reversible interfaces can fail to produce meaningful control when the design of choice itself is manipulative or unclear [30].

Other models address narrower weaknesses in static consent without resolving the broader problem. Retroactive consent is useful because it makes the timing of authorisation explicit, allowing consent and withdrawal to be analysed in terms of whether their effects apply to past or future data practices [33]. A related response is to reduce persistence prospectively by limiting the duration or openness of permission. Although Im et al. do not formalise ephemeral consent as a standalone governance model, their emphasis on consent as revertible and specific supports the idea that permission should be bounded and revisable rather than indefinite [17]. Yet these approaches remain partial. Once content has been copied, remembered, or redistributed by others, neither temporal expiry nor after-the-fact adjustment can fully restore sender control.

Seen comparatively, these models should not be treated as mutually exclusive competitors, but as attempts to repair different weaknesses in conventional consent. Dynamic Consent responds most clearly to temporality; Revocable Consent foregrounds withdrawal and reversibility; retroactive approaches sharpen how timing affects authorisation; and ephemeral approaches seek to limit persistence before problems arise. What unites them is a shared difficulty in accounting for messaging as a social practice in which the meaning and consequences of disclosure continue to evolve after the moment of sending [17, 22, 32, 33]. The central problem is therefore

not simply lack of choice, but the challenge of aligning revisability, revocability, relationality, and downstream loss of control within a single model.

For this reason, affirmative and ongoing consent offers the strongest conceptual direction for this dissertation. Im et al. describe affirmative consent as voluntary, informed, revertible, specific, and unburdensome, providing a framework that treats permission as active and revisable rather than presumed or exhausted at a single point in time [17]. These principles fit messaging more closely than one-off models because they better reflect how consent is negotiated in unfolding interaction. At the same time, they should not be overstated as a technical solution to screenshots, retention, or redistribution. Their value is conceptual and normative: they offer a clearer account of what responsible design should support, namely permission that can be revisited, clarified, limited, and renegotiated without assuming that post-sharing control can ever be complete.

This issue becomes even more complex in photo sharing, where consent may extend beyond a single sender-recipient relationship.

Taken together, the literature suggests that the main weakness of conventional consent models is not simply insufficient granularity, but their inability to account for the ongoing social life of shared content. Improved models repair important dimensions of this problem, but they still encounter hard limits once messages and photos enter interpersonal networks where recipients can retain, reinterpret, copy, or redistribute them. For everyday messaging, meaningful consent is therefore best understood not as a completed transaction but as an ongoing and socially situated process whose practical force depends on both relational norms and the technical limits of post-sharing control.

2.5 Consent in Instant Messaging and Photo Sharing

This section connects the consent concepts and theories to the specific realities of photo sharing in instant messaging, focusing on WhatsApp context.

2.5.1 Multi-Party Privacy Conflicts (MPCs)

Multi-party privacy conflicts arise when a single photo implicates several people whose privacy interests do not align, making individualised models of consent inadequate for photo sharing. In the photo-sharing literature, MPCs are generally understood as situations in which the person

who decides to share content is not the only person who bears the privacy consequences of that disclosure. One user's act of sharing can expose others depicted in the image, even where those others have had no meaningful opportunity to approve, refuse, or shape that disclosure. Cherubini et al. show that this problem is especially salient for co-owned content such as group photos, while Liu et al.'s survey of image privacy similarly suggests that privacy in online image sharing is shaped not only by image content itself but also by relationships, context, and multi-party interaction [8, 26]. Applied to instant messaging, this means that a shared photo may simultaneously involve the sender, the recipient, and any depicted co-subjects, each of whom may hold different expectations about whether the image should be sent, retained, forwarded, or later withdrawn. Consent in photo sharing is therefore not only ongoing, as argued in the previous subsection, but also distributed across multiple stakeholders from the outset, which makes sender-only models of permission normatively and practically incomplete [8, 26].

The literature proposes several responses to multi-party privacy conflicts, but these approaches often manage the problem more effectively than they resolve it. Cherubini et al. distinguish between precautionary mechanisms, which prevent or constrain sharing before disclosure, and dissuasion mechanisms, which seek to discourage risky sharing by prompting reflection on possible consequences [8]. They find that participants generally prefer precautionary mechanisms because these provide stronger protection for affected parties and reduce uncertainty for uploaders, although they also introduce friction and depend on identifying co-owners in advance [8]. This limitation is reflected in technical proposals. Xu et al. (2017) propose a system in which people appearing in a photo can be identified and involved in the sharing decision, but this depends on a privacy-preserving facial-recognition process to detect co-owners in the first place [48]. Again in another paper, Xu et al. (2019), similarly develop a trust-based privacy-preserving sharing model that allows selective protection of co-owned content, yet the approach still treats the problem largely through pre-sharing optimisation and does not fully incorporate direct negotiation among all affected parties [49]. Zhang et al. (2023) extend this critique by arguing that privacy-preserving frameworks often define ownership too narrowly and do not fully address the risks introduced by subsequent forwarding and cross-platform dissemination [50]. Taken together, these approaches are important because they show that co-ownership cannot be reduced to sender-side privacy settings alone, but they remain largely pre-sharing, system-centred, or prediction-based, and therefore struggle to reconcile competing interests once the image becomes socially and technically mobile [8, 48, 49, 50].

Multi-party privacy conflicts therefore show that consent revocation in messaging is not simply a matter of allowing one sender to change their mind, but a broader problem of shared stakes and weakening control once images begin to circulate [8, 50].

2.5.2 Persistence, Recipient Control, and the Limits of Revocation

Multi-party privacy conflicts show that control over a photo may already be shared before it is sent; the next limit is structural. In instant messaging, revocation does not operate as ongoing access control after delivery, because the system's basic function is to transmit content to a recipient-controlled device rather than to preserve sender authority over it indefinitely. For that reason, revocation is best understood as prospective and partial: once a photo has been delivered, a platform may still alter later in-app availability, but it cannot guarantee retrieval of the original disclosure. This is consistent with earlier work on revocation as a forward-looking rather than fully retroactive mechanism, and with Schnitzler et al.'s observation that deleting messages from a recipient's history is substantially harder than deleting them locally, even in closed messaging ecosystems [1, 32, 35].

WhatsApp provides a useful case because its privacy architecture and its revocation features point in different directions. On the one hand, WhatsApp emphasises end-to-end encryption and states in its UK Privacy Policy that messages are typically stored on users' devices rather than on its servers once delivery has occurred; on the other hand, this very model means that post-delivery control is necessarily limited [42, 46]. Deletion, disappearing messages, and view-once media should therefore be interpreted as bounded platform-level controls over later visibility, not as guarantees of erasure. WhatsApp's own help documentation states that recipients may see a message before deletion succeeds, that unsuccessful deletion may not be flagged to the sender, and that iOS recipients may still have media saved in Photos even after the message is deleted from the chat [44]. Likewise, disappearing messages only affect new messages and can be kept in the chat, while unopened view-once media can still be restored from backups; even newer protections such as Advanced Chat Privacy are described as helping prevent content from being taken outside WhatsApp by restricting export and auto-download, which reinforces that these features mitigate persistence routes rather than eliminate them [40, 41, 43, 45].

The weakness of revocation is not only technical but also interpretive. Schnitzler et al. show that users do not always correctly understand what a deletion action actually does, and that

clearer interface explanations improve their ability to assess deletion outcomes; in their study, participants in the WhatsApp condition understood message deletion more accurately than those in the Skype condition, partly because WhatsApp more clearly distinguishes local from broader deletion effects [35]. The same study also shows that deletion notices and other residual traces matter socially: participants were sometimes surprised not by whether deletion occurred, but by the notification that a message had been deleted, and some explicitly associated such traces with mistrust or ambiguity. In this sense, revocation in WhatsApp photo sharing is also an expectation-management problem. If interface cues suggest stronger control than the system can actually deliver, users may overestimate what deletion has achieved; if revocation is made too visibly traceable, it may also generate interpersonal friction rather than simply restore privacy [35].

If this subsection shows that revocation is already limited within the platform once a photo has been delivered, the next subsection shows that persistence becomes even harder to govern when recipients can capture and recirculate content through channels the platform cannot fully control.

2.5.3 The Screenshot Threat

The screenshot threat matters because it creates a form of persistence that no longer depends primarily on the platform's own retention settings. WhatsApp's privacy-oriented features, including disappearing messages, 'view once' media, and advanced chat privacy, can limit some later visibility or sharing within the app, but they do not fully govern what a recipient can do after viewing content [40, 41, 43, 46]. As Heath et al. note, even disappearing-message systems leave open preservation routes such as forwarding, screenshotting, copying, saving, or photographing content before it disappears [15]. As Ingber argues, screenshots turn part of a conversation into a new file on the recipient's device, making it easier to store, reuse, and circulate beyond the original exchange [18]. Screenshotting therefore creates a parallel persistence route that sits partly outside the logic of in-app revocation and ephemerality: even if the original message becomes less visible within WhatsApp, a copied representation may continue to exist elsewhere. More broadly, image-privacy scholarship identifies the recipient side as a continuing source of privacy risk once content is already in circulation, because recipients may retain or recirculate images beyond the sender's intended audience [26].

What makes screenshots especially important for this dissertation is that they are not only

technical workarounds but socially meaningful acts. Ingber distinguishes screenshot collection from screenshot sharing and shows that both can generate distinct privacy harms, because they move co-owned information into a new context without notice or consent [18]. This matters for consent because, as Im et al. note, affirmative consent must be specific: permission to share a photo with one recipient does not automatically extend to copying or recirculating it beyond the original interaction [17]. In WhatsApp photo sharing, screenshots therefore make revocation not only technically weak but normatively contested, as recipients may treat capture as harmless remembering, evidence, or legitimate reuse while senders may understand it as a breach of containment and trust.

For this reason, screenshot risk should be treated not as a marginal loophole but as a central challenge for any interface that presents revocation or ephemerality as meaningful user control [15, 18, 26].

2.6 Human Computer Interaction (HCI) and Interface Design

This section synthesises HCI research on why consent and revocation controls in digital platforms often fail in practice, and which interaction patterns can support meaningful, revisable user control in fast-paced messaging photo-sharing contexts [5, 47]. It focuses on the usability and integrity of control—whether users can locate, understand, and confidently act on privacy options without being nudged or steered toward platform-favoured outcomes [30].

2.6.1 Interface Failures and Manipulative Patterns

Even when platforms formally provide privacy and consent controls, HCI research shows these mechanisms often fail to produce meaningful user control at the interface level. This is an issue that becomes especially consequential in messaging-based photo sharing, where disclosures are often intimate, fast, and difficult to reverse once an image leaves the sender's device. Classic notice-and-choice approaches are frequently ineffective because notices are not usable or useful in practice, are often ignored, and are frequently treated as something bolted onto a system rather than integrated into its interaction design [34]. This breakdown is reinforced by broader critiques that modern data ecosystems create information asymmetries and complexity that make privacy self-management unrealistic: privacy policies are often too long and complex

to support real understanding, resulting in passive acceptance rather than informed decision-making [5, 47]. In messaging and social media settings, this becomes visible as concrete usability failures: expert evaluations using privacy-focused heuristics find that the ability to revoke consent is among the most problematic areas across popular apps, and that privacy information is inconsistently placed in menus in ways that undermine core usability principles such as consistency [2]. For photo sharing, the implication is straightforward: users cannot withdraw consent meaningfully if image-related controls (e.g., deletion, visibility, sharing restrictions) are hard to locate, hard to compare, and ambiguous about what they actually change—particularly when users are acting quickly in-chat and assume a "delete" action restores privacy.

This problem is intensified when interface designs do not merely confuse users but actively steer them toward platform-preferred outcomes. Empirical work on consent dialogs shows that "consent" interfaces commonly use design tactics that encourage acceptance of defaults (e.g., making "accept" more prominent, hiding configuration behind additional steps, or omitting an obvious decline path), so the banner functions more like a notification than a genuine control mechanism [4]. Analyses of GDPR-oriented consent pop-ups likewise argue that dark-pattern strategies can be used to circumvent the regulation's intent "by design," and that rights-protecting regulation is weakened if interface design is left unaddressed [38]. From a legal/HCI perspective, this steering is especially problematic because valid consent requires a meaningful choice and the ability to withdraw consent, and designs that make "accept" more salient than "reject" or that hide privacy-relevant options undermine that balance [13]. Experimental evidence also suggests that cookie-consent regimes often do not work as intended: participants frequently select privacy-unfriendly options and report low perceived control, raising concerns that interface-level nudges and defaults can hollow out the ideal of deliberative consent [14]. For WhatsApp-style photo sharing, the key implication is that revocation controls must not only exist but must be reachable, interpretable, and resistant to interface choices that overstate user agency or make withdrawal harder than acceptance.

Taken together, these failures motivate the next subsection's focus on what interaction patterns can actually support informed, revisable consent—for example, by treating notice timing and control presentation as explicit design dimensions [34] and by using interaction techniques that increase user attention to what they are disclosing [25].

2.6.2 Interactive Design for Meaningful and Revocable Consent

Constructive HCI work suggests that meaningful consent is best supported when it is designed as an explicit, understandable interaction rather than a frictionless checkbox event. In e-consent, Wilbanks argues that interface design shapes whether people actually attend to and understand what they are agreeing to, implying that consent should be treated as a designed experience that encourages comprehension rather than mere completion [47]. Relatedly, affirmative-consent work frames consent as requiring conditions such as being informed, specific, and revertible, which translates at the interface level into making actions intentional, consequences legible, and outcomes visible [17]. On mobile devices in particular, interaction techniques can be used to increase user engagement with the consent act itself: Lindegren et al. show that alternative smartphone consent designs intended to engage users (beyond a standard tap-through) can increase attention to what is being agreed to [25]. For messaging photo sharing, these insights imply that consent-relevant controls should be embedded into the moment of sending in a way users can notice under time pressure, and should provide clear feedback and states so users can tell what has changed after they act.

The same interactional logic extends from initial permission to later revision and withdrawal. Schaub et al. emphasise that the effectiveness of privacy notices depends on design dimensions such as timing and delivery context, supporting the use of just-in-time and contextually placed cues rather than relying on static policies [34]. For photo sharing, this points toward revocation controls that remain reachable after sending and are paired with explanations that honestly communicate scope. Where revocation is difficult to operationalise after disclosure, HCI work on multiparty privacy conflicts shows that design can intervene earlier through precautionary mechanisms (e.g., requiring collaboration/approval) or dissuasive mechanisms (e.g., friction or warnings that prompt reconsideration) [8]; these approaches are especially relevant to photo sharing because they can reduce avoidable disclosures before images become socially and technically mobile. Dynamic-consent literature strengthens this orientation by treating permissions as reviewable and updateable over time via interface-supported communication and preference management [22, 24, 36]. Technical proposals for dynamic consent management further highlight capabilities such as explicit expression of consent choices, access to consent history, and withdrawal support across a data lifecycle [27], while adoption-focused reviews emphasise that continuous communication and multiple modalities can facilitate use, but also note practical barriers such as consent fatigue and implementation complexity [24]. For WhatsApp-style

photo sharing, the design takeaway is not to promise perfect post-sharing control, but to build interactions that help users act deliberately at sharing time, understand the practical scope of revocation, and revisit decisions later through clear, contextual, and interpretable controls.

Overall, for messaging-based photo sharing the central HCI challenge is to design consent and revocation interactions that remain understandable and revisable over time, without implying that interface controls can fully overcome the limits of downstream retention and redistribution.

2.7 Empirical Usability Assessments of Current Platforms

Empirical evaluations suggest that revocation-related privacy controls are frequently available in principle but weak in usable effectiveness. In a STRAP-based expert heuristic evaluation of WhatsApp, Twitter, and Snapchat, Albesher and Alhussain identify the "ability to revoke consent" as the most notable problem area across all three apps, noting that none provides a clear way to revoke the underlying user agreement and that revocation (where offered) can be conditional or unreliable in its practical effect [2]. They also report that privacy information and settings are presented inconsistently across apps, undermining basic usability principles such as consistency and making control harder to locate and interpret [2]. Complementary legal-empirical evidence from the web consent ecosystem reinforces this availability–effectiveness gap: Kancherla et al. find that many popular sites make revocation interfaces absent or more complex than acceptance, and that revocation is often not correctly recorded or communicated to third parties, with tracking sometimes persisting even after users attempt to withdraw consent [21]. Taken together, these results support a general empirical takeaway: revocation tends to be a high-severity usability weakness, not simply a missing feature.

WhatsApp provides a concrete case of revocation-adjacent controls that remain bounded and context-dependent rather than fully reliable or granular consent management. Platform features such as message deletion, disappearing messages, view-once media, and chat-level protections can reduce some forms of persistence and re-sharing within the app, but they operate as mitigations rather than guaranteed erasure or downstream control [40, 41, 43, 44, 46]. Forensic analysis of ephemeral messaging further illustrates why these controls remain partial in practice: Heath et al. document how disappearing-message behaviour varies across platforms and storage locations, and they note WhatsApp's own acknowledgement that content can be preserved through actions such as replying (quoted content), forwarding into contexts where disappearing settings differ, or capturing content via screenshots/photography before expiry

[15]. Their results also show that message timers and residual artefacts can remain in application databases, reinforcing that ephemerality does not automatically eliminate persistence [15]. Overall, existing platforms provide controls that gesture toward revocation, but empirical and platform evidence suggests these controls remain partial, inconsistently interpretable, and weaker in practice than their presence implies.

2.8 Research Gap and Summary

Taken together, the literature reviewed in this chapter shows that consent revocation in messaging is neither a purely technical function nor a fully solved interface problem. Earlier sections establish that revocation is socially negotiated, only partially effective after delivery, vulnerable to forwarding and screenshots, and difficult to translate into interaction models that users can understand and act on in practice [15, 18, 30, 31, 32, 47]. The remaining research gaps therefore concern not whether revocation matters, but how it can be operationalised, scoped, and evaluated in WhatsApp-like photo-sharing contexts.

2.8.1 RG1: Translating dynamic/ongoing consent into usable messaging UI

The chapter has shown that dynamic, ongoing, and revocable consent are conceptually better suited to messaging than one-off authorisation, because they treat permission as revisable over time rather than exhausted at the moment of sending [17, 22, 24, 36]. However, the literature reviewed here provides much stronger conceptual and governance-level accounts than concrete guidance on how such models become low-friction, intelligible interaction flows within routine messaging. The unresolved issue is therefore one of translation: how ongoing consent can be expressed through usable controls, reviewable states, and in-context permission management for everyday photo sharing without overwhelming users or overstating what those controls can achieve [27, 47].

2.8.2 RG2: Post-delivery control is partial and easily misunderstood under real redistribution conditions

The chapter also shows that real platforms already offer deletion and ephemerality features, but that these operate only as bounded post-delivery mitigations rather than enforceable recall. In WhatsApp-like systems, revocation becomes structurally weaker once content reaches recipient-

controlled devices, and users do not always interpret deletion cues or feature scope accurately [32, 35, 41, 43, 44]. This problem intensifies under actual redistribution conditions, because forwarding, saving, screenshotting, and other capture practices create persistence routes beyond platform-scoped revocation [15, 18]. The remaining gap is therefore not simply that current controls are imperfect, but that there is still no robust model for post-delivery control that remains user-comprehensible once circulation extends beyond the original in-app exchange.

2.8.3 RG3: Multi-party privacy and fairness are under-supported in revocation interaction design

Across the chapter, privacy has been framed as relational, context-dependent, and frequently multi-party: once a photo is shared, senders, recipients, and depicted co-subjects can all have legitimate but competing stakes in how it is retained, circulated, or withdrawn [8, 26, 29, 31]. Yet current messaging tools largely encode revocation as a sender-centred action, offering little support for co-subject agency, negotiation, or explanations that make revocation intelligible to all affected parties. The unresolved issue is therefore one of fairness as well as control: the literature does not yet offer a clear interaction model for revocation that balances stakeholder visibility, minimises unnecessary interpersonal blame, and remains understandable in socially sensitive exchanges [8, 18].

2.8.4 RG4: Limited WhatsApp-specific empirical understanding of revocable photo-permission expectations

The reviewed literature gives a strong account of why WhatsApp is an important case: its controls are widely used, bounded in scope, and socially meaningful in ways that affect trust, interpretation, and downstream sharing [18, 35, 40, 44, 46]. It also provides platform documentation, heuristic usability evaluation, and evidence about screenshot or deletion effects [2, 15, 18, 35]. What remains under-evidenced, however, is what WhatsApp users themselves expect from revocable photo permissions in ordinary relational contexts: what forms of control they would understand, trust, accept, or regard as socially fair. The final gap is therefore empirical and context-specific, centring on user expectations rather than on the technical existence of platform features.

2.8.5 Summary

Together, these gaps show that the central problem is not simply whether revocation exists, but whether it can be translated into interaction models that are understandable, fair across stakeholders, realistic about post-delivery limits, and grounded in the expectations of WhatsApp users. They therefore motivate the need for a messaging-appropriate mechanism that supports usable, socially intelligible, and realistically scoped revocation of photo access in WhatsApp-like contexts.

Chapter 3

Requirements

3.1 Brief

This chapter defines the requirements for the WhatsApp-like proof-of-concept developed in this dissertation. Drawing on the literature review, it identifies the relevant stakeholders, use cases, functional requirements, and non-functional requirements for a prototype centred on sender-controlled photo permissions and revocation. The implemented scope includes pre-send and post-send permission configuration, recipient understanding of changing access states, constrained forwarding to a secondary recipient, and co-subject-related features within the current prototype. Together, these requirements translate the dissertation's broader privacy and consent concerns into a structured and implementable system specification. Broader issues that remain beyond the present prototype scope are identified separately as deferred concerns.

3.2 Stakeholders

1. **Photo Sender:** The sender shares the photo and configures its permissions. This stakeholder requires clear pre-send and post-send controls, visibility over the current state of a shared photo, and interaction patterns that allow revocation or permission change without unnecessary confusion.

2. **Primary Recipient:** The primary recipient receives the sender's photo in the original chat. This stakeholder requires clear feedback about whether a photo remains visible, or

whether any kind of revocation action has been taken.

3. **Secondary Recipient:** The secondary recipient appears only when the primary recipient forwards a photo. This stakeholder is necessary because the prototype examines whether sender-defined permissions continue to shape access after redistribution within the platform model.

4. **Co-Subject:** The co-subject is a person depicted in a photo. This stakeholder is necessary because the prototype examines whether co-subjects can request changes to the photo's permissions.

3.3 Prototype Scope (tekrar bakilacak)

The current functional prototype is intentionally narrower than the full conceptual problem described in the literature review. It implements sender-side configuration of photo permissions, sender-side revision of those permissions after sending, recipient-side feedback about active or revoked access, and a constrained forwarding flow to a secondary recipient. It demonstrates these mechanisms at the interface and interaction level rather than as a production-grade messaging infrastructure. It does not implement co-subject request workflows, co-subject conflict resolution, identity or face verification, persistent audit logging, full screenshot resistance, hard guarantees against saving or offline retention, and true cross-device enforcement beyond the local proof-of-concept model.

The following concerns are treated as out of scope for the present requirements set and are better positioned in discussion and future work chapters: co-subject agency, platform-assisted removal, identity or face verification, screenshot prevention, persistent audit logging, cross-device synchronisation, and guarantees of universal deletion after disclosure.

3.4 Use Cases

(UC-S0): Sender Configures Permissions Before Sending

1. The sender opens a chat and selects or uploads a photo.

2. The sender prepares the photo for sharing by configuring the following settings:

    - Forwaring permission

- View once permission
- Download permission
- Expiry Permission
- Co-subject declaration

3. The sender sends the photo into the chat.
4. The photo appears in the conversation with the chosen permission state applied.

(UC-S1): Sender Revises Permissions After Sending

1. The sender returns to a chat containing a previously shared photo.
2. The sender clicks the photo options button in the top-right corner of the photo message.
3. The sender selects 'Revocation permissions' from the in-chat action menu.
4. The system displays the available permission controls in the same in-chat menu.
5. The sender revises one or more permissions.
   - The sender revokes or restores photo access.
   - The sender allows or blocks forwarding.
   - The sender allows or blocks downloading.
   - The sender sets, changes, or removes a time limit.
6. The system updates the photo state in the chat.
7. The sender and other relevant participants see the revised access state.

(UC-R1): Primary Recipient Encounters Active, Revoked, or View-Once Access

1. The primary recipient opens the chat containing the sender's photo.
2. The recipient views the photo's current state in the chat interface.
3. The system presents the photo according to its current permission state, so that the recipient either sees the photo, opens it as view-once content, or is informed that access

is no longer available.

4. The recipient may open the photo options menu from the button in the top-right corner of the photo message.

5. The recipient acts within the limits of the available permissions, such as replying, forwarding, downloading, or requesting access.

(UC-R2): Primary Recipient Forwards an Allowed Photo

1. The primary recipient opens a chat containing a received photo.

2. The recipient attempts to forward the photo.

3. The system checks whether forwarding is currently permitted by the sender's settings.

   - If forwarding is allowed, the forwarding option remains available.

   - If forwarding is not allowed, the forwarding path is blocked.

4. The photo is forwarded with the sender-defined constraints preserved in the prototype model.

(UC-SR1): Secondary Recipient Receives Constrained Forwarded Content

1. The secondary recipient receives the photo in a separate forwarded chat after the primary recipient forwards it.

2. The system displays the forwarded photo using the original sender's current permission state.

3. The secondary recipient views the forwarded content through the interface.

4. The secondary recipient can interact with the photo only within the inherited constraints.

5. If the sender later revokes access or the photo expires, the forwarded content also becomes unavailable in the secondary recipient's chat.

(UC-C1): Co-Subject Requests Revocation of a Shared Photo

1. The co-subject opens their view of the chat containing a photo in which they are depicted.

2. The system shows that the co-subject appears in the shared photo.

3. The co-subject opens the photo or views its current access state through the interface.

4. The co-subject opens the photo options menu from the button in the top-right corner of the photo message.

5. The co-subject selects 'Request revocation'.

6. The co-subject may add an optional message explaining the request.

7. The system sends the revocation request to the sender for review.

3.5 Functional Requirements (FR)

ID	**Requirement**	**Purpose**	**Literature Support**	**Research Gap Addressed**
1. Pre-send consent configuration				
FR1	The prototype must allow the sender to select a photo and configure permissions before sending.	Provide a pre-send composer that lets the sender set photo permissions; forwarding, view-once behaviour, time limits, co-subject declaration, and saving/export permissions before sending.	Dynamic, revocable, and ephemeral consent; in-context permission at send. [8, 15, 17, 22, 24, 33, 34, 36, 40, 41, 43, 47]	RG1

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
FR2	The prototype must explicitly address screenshot and download risk (e.g. warn before sharing, indicate that screenshots cannot be fully prevented).	Treat screenshot risk as central even where perfect prevention is impossible.	Treat screenshot risk not as marginal loophole but as central challenge; honestly communicate scope; warnings that prompt reconsideration. [8, 15, 18, 34, 40, 41, 43, 46]	RG2
2. Post-send revocation and permission management				
FR3	The prototype must allow the sender to modify or revoke access to a previously shared photo via contextual in-chat, photo-level controls (e.g. action menu with reply, forward, save/export, and a revocation-permissions submenu).	Provide post-send controls via a photo-level action menu so the sender can change visibility, forwarding permission, expiry, and saving/export for an existing photo message.	Controls that remain reachable after sending; permission revisable over time; withdrawal support. [1, 2, 27, 32, 34, 35]	RG1 and RG2

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
FR4	The prototype must show all users clear states for active, revoked, view-once access and expiry duration.	Render recipient-facing feedback that distinguishes visible content, revoked content, and single-view media within the chat interface.	Users do not always understand what revocation does; clearer cues improve comprehension; legible consequences and visible outcomes. [2, 17, 34, 35, 41, 43, 44, 47]	RG2
FR5	The prototype must maintain a live consent state per shared photo recording current permission status, expiry, active or revoked access, and applicable restrictions.	Provide a usable basis for later revocation and review; without this, revocation or review is not feasible.	Permissions as reviewable and updateable over time; explicit consent choices and consent history; revocation as closing access interval. [22, 27, 33, 47]	RG1

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
FR6	The prototype should provide clear feedback after key permission-related actions.	Make the outcome of sending, forwarding, or changing permissions understandable to the user.	Visible and interpretable feedback after privacy actions reduces ambiguity about what changed. [2, 34, 35, 47]	RG1 and RG2
3. Redistribution and forwarding control				
FR7	The prototype must constrain forwarding according to the sender's current permission state.	Allow forwarding only when the sender has enabled redistribution and block the forwarding path otherwise.	Forwarding changes information flow; redistribution should follow sender-defined transmission constraints; downstream sharing is a central privacy risk.[8, 15, 18, 26, 29, 40, 45, 50]	RG2

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
FR8	The prototype must propagate sender-defined permissions to forwarded content within the proof-of-concept model.	Ensure that a forwarded photo shown to the secondary recipient inherits the original sender's permissions as represented by the interface.	Preserve original transmission principles across forwarding; inherited constraints maintain context and reduce downstream privacy loss.[8, 15, 26, 29, 40, 45, 50]	RG2 and RG3
FR9	The prototype should make the forwarded context and secondary-recipient view clearly interpretable.	Use separate chat presentation, forwarded labelling, or explanatory interface cues so that the secondary-recipient flow is understandable during evaluation.	Make redistribution state legible; forwarding changes audience and context, so secondary-recipient status should be clearly indicated. [2, 8, 26, 29, 34, 35, 47, 50]	RG2
4. Recipient Awareness and negotiated consent				

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
FR10	The prototype must provide recipients with a neutral inline indication when access is revoked or when permissions change.	Keep the recipient informed whilst using socially careful wording, given that revocation can create interpersonal tension.	Revocation as expectation-management; notices can generate mistrust or ambiguity; design for face-saving and minimal interpersonal blame. [8, 10, 18, 19, 31, 35, 37, 47]	RG3
FR11	The prototype must allow the recipient to request extended access, with the sender deciding whether to approve or reject that request.	Support consent as socially negotiated rather than only technically toggled.	Revocation as re-coordination of privacy boundaries; consent negotiated in unfolding interaction; permission revisited, clarified, renegotiated. [9, 17, 22, 24, 31, 36]	RG3
5. Co-subject and multi-party privacy control				

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
FR12	The prototype must support multi-party photos by allowing the sender to mark that a photo contains other people and allowing depicted co-subjects to request revocation of a shared photo, including an optional explanation for the request.	Reflect the literature's emphasis on multi-party privacy conflicts rather than assuming only the sender matters.	Multi-party privacy conflicts; depicted others with no opportunity to approve or shape disclosure; support for co-subject agency and shared stakes. [5, 8, 26, 29, 31, 48, 49]	RG3 and RG4
FR13	The prototype should allow a depicted co-subject to receive a notification that they appear in a shared photo when the sender has marked the image as multi-party content.	Make co-subject involvement visible and support awareness that a privacy-relevant disclosure has occurred.	Co-subjects often lack opportunity to approve or shape disclosure; awareness is a prerequisite for agency. [5, 8, 26, 48]	RG3

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
FR14	The prototype should allow the sender to review and respond to a co-subject's request through a dedicated decision interface.	Demonstrate how multi-party privacy could be operationalised through negotiation rather than unilateral hidden system logic.	Precautionary/collaborative mechanisms; privacy negotiation among stakeholders; consent as revisable and relational. [8, 17, 31, 48, 49]	RG3
FR15	The prototype should provide neutral feedback when a co-subject request leads to changed permissions.	Communicate changes clearly while minimising blame and interpersonal friction.	Revocation notices are socially meaningful; design should support face-saving and expectation management. [18, 31, 35, 47]	RG3
FR16	The prototype should show that co-subject-related permission changes can affect forwarding and secondary-recipient access within the platform model.	Demonstrate that co-subject privacy concerns continue after redistribution and are not limited to the original chat.	Forwarding changes context and audience; downstream dissemination intensifies multi-party privacy risk. [8, 26, 29, 50]	RG3

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
FR17	The prototype should allow the interface to explain the limits of co-subject protection, including that offline capture, screenshots, or prior viewing cannot be fully undone.	Keep co-subject controls realistic and avoid overstating the effect of platform-level revocation.	Revocation is prospective and partial; screenshotting and retention remain structural limits. [15, 18, 32, 35]	RG2 and RG3

Table 3.1: Functional Requirements

3.6 Non-Functional Requirements (NFR)

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
Usability				
NFR1	The interface must be easy to understand for first-time users.	Ensure that users can understand the permission model and core controls immediately without prior training or repeated use.	First-use interpretability; consent controls should be usable, legible, and consistently presented rather than hidden or ambiguous. [2, 5, 17, 30, 34, 35, 47]	RG1 and RG2

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
NFR2	The system shall support quick completion of routine sharing tasks whilst introducing only proportionate friction for privacy-sensitive actions; enough to support reflection, but not so much that users avoid the feature.	Balance everyday efficiency with reflective intervention so privacy-sensitive actions receive attention without making the system feel burdensome or disruptive.	Consent interactions should be unburdensome but reflective: routine actions should stay low-friction, while privacy-sensitive sharing can justify proportionate prompts or warnings that support attention without causing fatigue or avoidance. [3, 8, 17, 24, 25, 34, 47]	RG1

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
NFR3	Consent and privacy controls must present choices in a clear, balanced, and non-manipulative way.	Ensure users can understand and compare options without hidden choices, unequal emphasis, or pressure toward a preferred action.	Meaningful consent is undermined by dark patterns, unequal salience, hidden decline paths, and unclear notices; privacy controls should remain interpretable and support freely made choices. [4, 13, 30, 34, 38, 47]	RG1 and RG3
NFR4	The interface must provide readable and distinguishable permission states, labels, and controls.	Make permission-related information easy to read, compare, and interpret so users can recognise states and act confidently.	Permission states and controls must be visually legible and clearly distinguishable so users can interpret privacy status changes, locate actions, and understand consequences without ambiguity. [2, 17, 34, 35, 47]	RG1 and RG2
Performance				

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
NFR5	Permission-related actions and status changes should remain responsive enough for live demonstration in a local browser environment.	Preserve a smooth and credible interaction flow by ensuring that permission changes appear promptly during use and evaluation.	Low-friction interaction is important in fast-paced messaging; consent controls should remain usable at the moment of action and not introduce disruptive delay. [2, 17, 24, 25, 34, 47]	RG1
Social Impact				
NFR6	The interface must use neutral, non-blaming language to reduce unnecessary social tension when permissions change.	Reduce avoidable interpersonal friction by ensuring that permission changes are communicated in a socially careful and non-accusatory way.	Revocation is a socially meaningful boundary change, so interface wording should minimise blame, ambiguity, and interpersonal friction while keeping changes intelligible. [9, 10, 19, 31, 34, 35, 37, 47]	RG3
Consistency				

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
NFR7	Permission menus, indicators, and states should remain consistent with familiar WhatsApp-like interaction patterns.	Improve learnability and reduce confusion by aligning new permission controls with interaction patterns users already recognise from WhatsApp-like messaging.	Consistency supports interpretable privacy control: revocation-related settings should follow familiar messaging patterns and avoid inconsistent placement or presentation that makes actions harder to locate or understand. [2, 17, 34, 35, 41, 43, 44, 47]	RG1 and RG4

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
NFR8	Permission management should assume continuity with WhatsApp's existing end-to-end encryption model, without requiring changes to the underlying encryption architecture.	Preserve conceptual compatibility with users' expectations of secure WhatsApp-like communication while introducing permission controls.	This assumption ensures that the proposed permission model is interpreted as operating alongside, rather than replacing or weakening, WhatsApp's existing end-to-end encryption. It supports consistency by framing permission management as an additional layer of user control that remains compatible with familiar secure messaging expectations.	RG1 and RG4
Transparency				

ID	Requirement	Purpose	Literature Support	Research Gap Addressed
NFR9	The system must not over-promise. It should clearly indicate that revocation may not affect screenshots, saved copies, or prior viewing.	Calibrate user expectations by communicating the practical limits of revocation clearly and honestly.	Revocation in messaging is only partial and prospective, so the interface should communicate scope honestly and avoid implying complete erasure where screenshots, saved copies, exports, or prior viewing remain outside full platform control. [15, 18, 32, 34, 35, 40, 41, 43, 44, 46]	RG2

Table 3.2: Non-Functional Requirements

3.7 Feasible Extensions Consistent with Current Scope

If further implementation time is available, the most suitable additions are small interface-level improvements that strengthen the current revocation story without expanding the project into a different problem domain. The strongest candidates are an optional sender explanation for permission changes, clearer confirmation feedback after revocation events, and more explicit

labelling of forwarded content on the third phone. These additions would support evaluation and dissertation coherence without reintroducing the larger co-subject or infrastructure-heavy issues that are intentionally outside the current prototype scope.

Chapter 4

Design & Specification

4.1 Brief

This chapter presents the design and specification of the proposed consent revocation system for photo sharing in WhatsApp. It translates the requirements, use cases, literature findings, and platform constraints established in earlier chapters into a justified interaction design and system specification. The chapter first introduces the Smart Photos concept and design rationale, before covering the specification in section4.4.

4.2 Design Rationale and Principles

The design of **Smart Photos** is grounded in the central argument of this dissertation: consent in messaging-based photo sharing should not be treated as a one-off permission granted at the moment of sending, but as an ongoing and revisable process shaped by context, relationships, and the practical limits of post-delivery control. In the proposed model, each shared photo is treated not simply as delivered content, but as a photo associated with a persistent permission state that can specify whether it may be forwarded, saved, viewed once, or accessed for a limited period. These permissions are not exhausted at disclosure, but remain reviewable and revisable through the interface after sending. The design principles set out below are informed by Communication Privacy Management, Contextual Integrity, revised consent models, and privacy-focused HCI research [9, 17, 29, 31, 34, 47]. They provide the conceptual basis for the structure of the model and the prioritisation of particular interaction choices.

4.2.1 Dynamic and Revocable Consent by Default

The first design principle is that consent should be treated as dynamic and revocable by default rather than as a fixed decision exhausted at the point of disclosure. This follows directly from the literature reviewed in Chapter 2, which shows that one-off consent models are poorly suited to interpersonal messaging, where disclosures unfold over time and may later feel inappropriate as contexts, relationships, or expectations change [17, 22, 24, 36]. In CPM terms, sharing a photo does not eliminate the need for later boundary management; rather, it creates a situation in which privacy rules may need to be revisited and renegotiated after disclosure [9, 31].

For that reason, the design does not treat photo sending as an irreversible transfer of control. Instead, it is structured around the idea that a photo can remain associated with a modifiable permission state, allowing the sender to determine conditions of access before sending and to revise those conditions afterwards. This principle underpins the decision to support both pre-send configuration and post-send permission management in the prototype, rather than limiting user control to a single moment of choice. It also supports the inclusion of time-bounded access, view-once behaviour, and later permission revision as core parts of the design rather than optional add-ons. That also means a photo keeps its original sharing conditions after it is sent, so later edits or allowed redistribution within the prototype do not change the original permissions. In this way, the prototype operationalises the dissertation's first research gap by translating the literature on dynamic and ongoing consent into a messaging-oriented interaction model that remains reviewable after disclosure.

4.2.2 Expectation Management over Absolute Control

The second principle is that the interface should communicate realistic and bounded forms of control rather than implying that revocation can fully undo disclosure. The literature reviewed in Chapter 2 shows consistently that revocation in messaging environments is prospective and partial: once content has been delivered, viewed, copied, downloaded, or captured through screenshots or external devices, no interface can guarantee complete retrieval or universal deletion [15, 18, 32, 35]. This limitation is especially important in WhatsApp-like systems, where platform-level controls can affect later in-app visibility but cannot fully govern what has already been retained on recipient-controlled devices [41, 43, 44, 46].

This principle therefore rejects any design framing that would overstate the application's power. Within the **Smart Photos** model, revocation is treated primarily as a mechanism for modifying

or withdrawing future platform-mediated access and redistribution, rather than as a guarantee of complete deletion after delivery. This makes expectation management a core design responsibility rather than a peripheral disclaimer. It justifies the inclusion of screenshot and capture warnings, state explanations, and wording that emphasises limited but meaningful control. In practical terms, the goal is to help users form a more accurate mental model of what permission changes can and cannot achieve.

4.2.3 (suna bi daha bak) Minimal Social Friction

A third design principle is that revocation and permission change should be communicated in a way that minimises unnecessary interpersonal tension. The literature reviewed earlier shows that revocation is not only a technical event but also a socially meaningful act. Under CPM, changing access after disclosure may be interpreted as a renegotiation of privacy boundaries, while work on social presence and media richness suggests that changes to communication permissions can carry relational signals such as mistrust, distancing, or suspicion [10, 19, 31, 37]. Empirical work on deletion cues further suggests that visible traces such as deletion notices may themselves become socially consequential, sometimes producing ambiguity or discomfort rather than simply clarifying what happened [35].

For that reason, the design aims to reduce friction without hiding the fact that a permission change has occurred. This means using neutral, system-mediated wording and avoiding interface patterns that imply blame. The aim is not to hide revocation, but to present it in a way that it limits the possibility of unnecessary tension. This principle is particularly important for recipient-facing notices and status changes, where the interface must remain clear about changing access while avoiding designs that frame revocation as punitive or suspicious.

4.2.4 Fairness Between Parties

Although the project prioritises sender-side control, the design rationale does not treat consent management as a purely sender-centred matter. Both CPM and Contextual Integrity suggest that once content has been shared, the expectations of other parties also matter, even where the sender retains a legitimate interest in revising access [29, 31]. A sender may view revocation as a reasonable attempt to reassert privacy boundaries, while a recipient may interpret the same action as a change to previously granted access. This tension does not mean the sender should lose the ability to revise permissions, but it does mean that recipient-facing states and notices

must be understandable. Although the immediate interaction in the prototype centres on the sender and both types of recipients, the same concern with fairness also extends to photos that involve depicted co-subjects whose interests may be affected by later access or redistribution.

The design therefore seeks to balance sender agency with recipient understanding. Recipients are not treated as passive endpoints who simply lose access without explanation. Instead, the system provides readable states, neutral notices, and clear indications of what remains available, what has changed, and what actions are no longer permitted. It also supports limited request-based interaction, allowing a recipient to ask for extended or renewed access while leaving the final decision with the sender. In this way, the design reflects the literature's view of consent as a socially negotiated process and responds to the need for revocation mechanisms that remain socially intelligible rather than merely technically enforceable.

4.2.5 (bu tamam) Consistency with WhatsApp's Existing Interaction Patterns

The final principle is that new revocation and permission features should remain consistent with the interaction patterns users already associate with WhatsApp-like messaging. HCI research reviewed in Chapter 2 shows that privacy controls often fail not only because they are absent, but because they are inconsistently presented, or difficult to interpret in fast-paced use [2, 34, 47]. In messaging contexts, where actions are often taken quickly and with limited deliberation, recognisable interaction patterns are especially important for first-use clarity and adoption.

For that reason, the design does not attempt to invent an entirely separate privacy-management environment detached from ordinary messaging behaviour. Instead, it follows familiar WhatsApp-like interaction conventions such as photo-level action menus, recognisable icons, in-chat state cues, and lightweight bottom-sheet style controls. This principle supports the non-functional requirements concerned with learnability, clarity, and consistency, and helps ensure that the proposed system feels like a plausible extension of an existing messaging platform rather than an unfamiliar parallel application. By embedding consent and revocation controls within established patterns of use, the design aims to support meaningful control without introducing so much novelty or friction that the feature becomes difficult to adopt in everyday communication.

4.3 Conceptual System Model: Smart Photos

This section introduces the conceptual model that structures the prototype. The **Smart Photos** model treats each shared photo as an object associated with a persistent permission state, rather than as content whose control is fully transferred at the moment of delivery. Under this model, a photo is shared together with explicit permissions and constraints, such as whether it can be forwarded, saved, viewed once, or accessed for a limited period. These permissions are not fixed permanently at the point of sending. Instead, they remain attached to the photo as an ongoing state that can later be reviewed and revised through the interface.

This model is intended to support the dissertation's central view of consent as ongoing and revisable rather than exhausted at disclosure. It provides the conceptual basis for pre-send configuration, post-send permission management, recipient-facing state changes, constrained forwarding within the prototype, and cases where a shared photo may also involve co-subject interests. The model is intended to preserve the conditions under which a photo is originally shared, even if those conditions are later revised or the photo is redistributed within the platform. This means that permissions remain attached to the photo as it moves between users, rather than being reset each time it is shared.

A key position taken in this design is that consent revocation in WhatsApp photo sharing cannot be treated as a guarantee of complete deletion once a photo has been delivered to another user's device. Instead, the **Smart Photos** model conceptualises revocation primarily as a mechanism for modifying or withdrawing future platform-mediated access and redistribution. This framing follows directly from the constraints identified in Chapter 2, where messaging platforms such as WhatsApp were shown to operate within structural limits created by prior viewing, screenshots, local retention, and external capture. Treating revocation as equivalent to universal deletion would therefore risk producing misleading expectations and inaccurate mental models of control. The model instead aims to support meaningful but realistically bounded control by making permissions persistent, revisable, and visible, while remaining honest about the limits of what the system can enforce after disclosure.

4.4 Functional Specification

4.4.1 Pre-send Permission Configuration

To achieve FR1 and FR2, the prototype places photo-permission controls directly within the sender's sharing flow rather than treating privacy as something that can only be addressed after a photo has already been sent. From the sender's chat interface, the user can open a photo picker, choose an image from preset media, device files, or camera input, and then either send it immediately with default settings or open a dedicated composer through 'Send with Revocable Permissions' for permission-based sharing. This keeps permission decisions close to the moment of disclosure while preserving a lightweight path for routine sharing.

When the sender chooses to send a photo with revocable permissions, the prototype opens a dedicated pre-send composer. Within this composer, the sender can define the photo's initial permission state before it enters the chat. The current implementation allows the sender to set whether forwarding is allowed, whether the photo should be limited to view-once access, whether downloading is allowed, whether the photo should be marked as containing a co-subject, and whether access should expire after a user-defined time interval. Time-limited access is configured through a duration selector for days, hours, minutes, and seconds, and the sender can also add a caption before sending. In this way, the act of sending does not function as a simple transfer of content, but as the creation of a photo message with an attached permission state from the outset.

The prototype also supports a lower-friction direct-send path from the same picker. In that case, the selected photo is sent immediately using default settings rather than through the full composer. This allows the design to distinguish between routine sharing and more privacy-sensitive sharing, while still making richer permission control available at the point of send when the sender chooses it.

To achieve FR2, the pre-send flow also includes an explicit risk cue that communicates the limits of later control before disclosure takes place. In the current prototype, the composer shows a short warning that screenshots and saved copies cannot be undone. This does not attempt to prevent screenshotting or local retention technically; instead, it frames such risks as structural limits of platform-level control and makes them visible before the photo is shared. As a result, the pre-send interaction supports more informed permission-setting by combining configurable restrictions with an explicit reminder that later revocation cannot fully reverse all

forms of recipient-side capture or retention.

4.4.2 Post-send Permission Management

To achieve FR3, the prototype keeps permission management reachable after a photo has already been delivered through contextual, photo-level controls within the chat. Each sent photo includes a small action button in the top-right corner of the message bubble. Selecting this control opens an in-chat menu from which the sender can access both general photo actions and a dedicated revocation-permissions view. Through these post-send controls, the sender can revoke or restore visibility, allow or block forwarding, allow or block downloading, and add or remove a time limit for an existing photo. In this way, post-send control is treated as an ongoing interaction tied to the individual photo message rather than as a one-off decision made only at the moment of sending.

To achieve FR5, these changes operate on a live consent state associated with each shared photo. In the current prototype, this state records whether access is active or revoked, whether forwarding is permitted, whether downloading is permitted, whether the photo is marked as view-once, whether a co-subject flag is present, and whether access expires at a specified time. Because these permissions remain attached to the same source photo after sending, the prototype can update the state of an existing message rather than replacing it with a new one. The adjacent permissions panel reinforces this model by showing the currently selected photo together with its current status, forwarding condition, view-once state, download condition, and time limit, allowing the sender to inspect the active state before making further changes.

To achieve FR4, the prototype also renders these permission states clearly within the chat interface for all relevant participants. Beneath each photo, the interface displays compact state labels indicating whether the photo is visible or revoked, whether forwarding is allowed, whether the content is view-once or reviewable later, whether downloading is allowed, and, where applicable, how long remains before expiry. When access is revoked, the recipient-facing interface replaces the visible photo with an inline placeholder indicating that the photo is no longer available in the current chat. When a photo is configured as view-once, recipients open it in a full-screen view that becomes unavailable after closing. These interface states make permission changes legible in the conversation rather than leaving them implicit or hidden.

To achieve FR6, key permission-related actions are accompanied by lightweight but persistent feedback within the same interface. After sending, the sender is shown a short summary of

the most recent send action, and forwarded activity is also summarised separately in the side panel. More generally, the outcome of permission changes is reflected immediately through updated state chips beneath the photo, the current-photo status shown in the permissions panel, and any visible change in recipient-facing presentation, such as revoked placeholders or expiry countdowns. Rather than relying on detached confirmation dialogs, the prototype makes the results of post-send actions understandable by updating the same chat and control surfaces through which the photo is already being managed.

4.4.3 Recipient Experience and Permission Changes

To achieve FR10, the prototype provides the recipient with neutral inline indications when access is no longer available or when the current permission state has changed. In the recipient view, each photo message is accompanied by visible state labels showing whether the photo is currently visible or revoked, whether forwarding is allowed, whether the content is view-once or reviewable later, whether downloading is allowed, and, where relevant, how long remains before expiry. If access is revoked, the visible photo is replaced by an inline placeholder stating that the photo is no longer visible in the current chat. The wording used in these states remains system-like and descriptive rather than accusatory, so that the recipient is informed of a change in access without the interface framing it as blame or misconduct.

In the current prototype, this neutral recipient-facing feedback is delivered primarily through updated in-chat state labels, revoked-state placeholders, and short explanatory text beneath the photo message, rather than through a separate system notice for every permission change. This keeps permission changes visible within the same conversational context in which the photo was shared, while supporting the dissertation's aim of making revocation legible without introducing unnecessarily confrontational deletion-style cues.

To achieve FR11, the prototype also allows the recipient to request renewed access when a previously shared photo is no longer available. In the current implementation, when the recipient opens the action menu for a revoked photo, a request-access option becomes available. Selecting this opens a small request flow in which the recipient can send an access request to the sender together with an optional explanatory message. The request does not automatically restore access; instead, it is routed to the sender side of the prototype as a pending request associated with the source photo.

The sender can then explicitly accept or reject that request. If the sender accepts, the prototype

restores the photo's access state for the corresponding source photo. If the sender rejects, the recipient is shown a short follow-up message indicating that the access request has been rejected. In this way, the prototype treats access renewal not as an automatic technical reversal, but as a negotiated interaction in which the recipient can ask for renewed access while the sender retains control over the final decision.

4.4.4 Forwarding and Redistribution Control

To achieve FR7, FR8 and FR9, the prototype treats forwarding as a permission-dependent action rather than as an unrestricted continuation of ordinary message sharing. A photo can only be forwarded when the sender's current permission state allows redistribution and the photo remains active. In the chat interface, this is reflected through the photo-level action menu: when forwarding is blocked, the forwarding option is disabled; when it is permitted, the photo can be forwarded into the secondary-recipient chat. In this way, redistribution is constrained by the sender's current permission state rather than by recipient discretion alone.

To achieve FR8, forwarded content does not create a separate copy with an independent permission model. Instead, the forwarded message remains linked to the original photo through the same source-photo identifier and inherits the sender-defined permissions as represented in the interface. In the current prototype, this means that forwarded content continues to reflect the original photo's visibility status and other active restrictions, including forwarding availability, download permission, view-once behaviour, and any time-based expiry. If the sender later changes the permission state of the original photo, those changes are also reflected in the forwarded instance associated with that same source photo. The prototype therefore models forwarding as continued circulation under the original permission logic rather than as a break from the initial sharing conditions.

To achieve FR9, the prototype presents the forwarded flow as a distinct recipient context rather than leaving redistribution implicit. Forwarded content appears in a separate secondary-recipient chat, making the audience change visible during evaluation and helping distinguish the original recipient's conversation from the downstream one. Within that forwarded view, the interface also indicates that restrictions from the original chat still apply, reinforcing that redistribution remains governed by the sender-defined permission state. By separating the forwarded conversation while preserving inherited restrictions, the prototype makes downstream sharing legible as an extension of the original disclosure rather than as an unrelated or uncontrolled

disclosure event.

4.4.5 (bu silinecek)Co-Subject and Multi-Party Privacy Control

To achieve FR12 and FR13, the prototype extends the permission model beyond the sender/recipient pair by allowing the sender to mark, at send time, that a photo contains a co-subject. In the current implementation, this is done through the pre-send composer, where the sender can enable a co-subject flag as part of the photo's initial permission state. Photos marked in this way then become visible within a dedicated co-subject view, where the depicted person is shown a clear indication that they appear in the photo. Rather than treating the image as sender-only content, the prototype therefore represents certain photos as multi-party disclosures whose privacy implications extend beyond the original sender and recipient.

To achieve FR13 more specifically, the co-subject is not only shown the marked photo but is also given an explicit interface cue stating that they are depicted in it. In the current prototype, this appears as a dedicated co-subject card with a short notification message indicating that the person appears in the shared photo and may request revocation. This makes co-subject involvement visible in the interface and provides a basis for later action rather than leaving multi-party implications implicit or hidden.

To achieve FR12 and FR14, the prototype also provides a co-subject request flow through which the depicted person can ask the sender to revoke the photo. This request is initiated from the co-subject's photo-level action menu and opens a small modal in which the co-subject can send a revocation request together with an optional explanatory message. The request is then routed to the sender side of the prototype as a pending item associated with the relevant source photo. Rather than applying revocation automatically, the sender is presented with an explicit accept-or-reject decision interface in the sender-only message area. This keeps co-subject privacy management within a negotiated interaction model rather than reducing it to hidden system behaviour.

To achieve FR15, the outcome of a co-subject request is communicated through clear but non-blaming feedback. On the sender side, accepted and rejected requests are shown with short outcome labels indicating whether the photo has been revoked or the co-subject has been notified of rejection. If the sender rejects the request, the co-subject receives a follow-up message stating that the revocation request has been rejected. If the sender accepts, the photo's access state is updated without framing the change in accusatory terms. In this way, the prototype

makes the outcome of co-subject intervention visible while still aiming to minimise unnecessary interpersonal friction.

To achieve FR16, co-subject-related permission changes continue to apply beyond the original chat. In the current prototype, accepting a co-subject revocation request changes the permission state of the original source photo itself. Because forwarded content remains linked to that same source photo and inherits its effective permission state, a co-subject-triggered revocation also affects the forwarded instance shown to the secondary recipient. The interface therefore models co-subject privacy as extending into downstream redistribution contexts rather than stopping at the original sender/recipient exchange.

To achieve FR17, the prototype also makes the limits of co-subject protection explicit. In the co-subject view, the interface states that protection is in-app only and that screenshots and prior viewing cannot be fully undone. The revocation-request flow similarly reminds the co-subject that offline copies may already exist. These cues ensure that co-subject controls are presented as meaningful but bounded forms of platform-level control, rather than as guarantees of complete erasure after disclosure.

4.5 Non-Functional Specification

4.5.1 Usability and Interaction Clarity

To achieve NFR1, the prototype presents permission-related controls within familiar chat and media-sharing interaction points rather than requiring users to learn a separate privacy-management environment before use. Core actions are embedded directly into the sender's chat interface through the photo picker, the pre-send composer, the in-chat photo menu, and the adjacent permissions panel. This is intended to make the permission model understandable from first use by keeping controls close to the photo they affect rather than hiding them in detached settings screens or abstract settings menus.

To achieve NFR2, the interface supports quick completion of routine sharing tasks while introducing additional interaction only where privacy-sensitive reflection is useful. In the current prototype, a selected photo can either be sent directly with default settings or opened in a dedicated composer for permission-based sharing. This preserves a lightweight path for ordinary sharing, while still allowing the sender to configure forwarding, view-once behaviour, download permission, co-subject marking, and time-limited access before disclosure when more deliberate

control is needed. In this way, the prototype aims to introduce proportionate friction rather than treating all sharing actions as equally sensitive.

To achieve NFR3, consent and privacy choices are presented as explicit, interpretable options rather than as hidden or one-sided defaults. In the pre-send composer, permission controls are shown as individually labelled buttons within the same visible panel, and the sender can either enable or leave each option disabled without navigating through obscured submenus or being forced toward a single preferred configuration. Similarly, post-send actions such as revoking access, blocking forwarding, blocking download, or adding a time limit are presented as explicit menu choices within the photo-level action flow. This keeps the interface relatively transparent about what options are available and avoids structuring privacy control as a hidden or manipulative interaction.

To achieve NFR4, permission states, labels, and controls are designed to remain visually distinguishable across the interface. The prototype uses clearly separated action menus, labelled permission buttons, compact state indicators beneath photo messages, and distinct placeholders for revoked content so that users can recognise the current state of a photo without having to infer it indirectly. This applies both to sender-side management and to recipient- and co-subject-facing views, where visible changes in access, forwarding, download permission, view-once status, and expiry need to remain legible after disclosure.

To achieve NFR5, permission-related actions remain responsive within the local browser-based prototype so that changes in state appear immediately during interaction and demonstration. In the current implementation, permission updates are applied directly to the active photo state and are reflected at once in the chat view, the permissions panel, forwarded content, and recipient-facing or co-subject-facing states where relevant. This helps preserve a credible and continuous messaging flow, particularly when the sender revises permissions after sending and the resulting changes need to remain immediately visible to other participants. Together, these choices specify usability and interaction clarity as core qualities of the prototype rather than as secondary presentation concerns.

4.5.2 Socially Careful Communication

To achieve NFR6, the prototype communicates permission changes through neutral, system-mediated wording that avoids unnecessary blame or accusation. This is especially important because revocation, restriction, and access renewal are not only technical actions but also

socially meaningful changes in interpersonal access. In the current interface, permission changes are therefore phrased in descriptive terms such as a photo no longer being visible, access having changed in the current chat, or a request having been accepted or rejected, rather than in language that implies wrongdoing on the part of the recipient or frames the sender's action as punitive.

This principle is reflected in both photo-state presentation and request outcomes. When access is revoked, the recipient sees an in-chat placeholder stating that the photo is no longer visible, supported by short explanatory text and updated state labels. Similarly, request-based interactions are communicated through restrained feedback such as indicating that access has been restored, that a revocation request has been accepted, or that a request has been rejected and the affected party has been notified. These messages make the outcome of permission-related actions clear without relying on confrontational or suspicion-laden deletion-style cues.

In this way, socially careful communication is specified as a core non-functional quality of the prototype: permission changes must remain visible and understandable, but they should do so in language and interface forms that minimise avoidable interpersonal friction.

4.5.3 Consistency, and Honest Communication of Limits

To achieve NFR7, the prototype keeps permission menus, indicators, and message states consistent with familiar WhatsApp-like interaction patterns. Permission controls are embedded within recognisable chat elements, including the photo picker, the pre-send composer, the in-chat photo action menu, the sender-side permissions panel, and the state labels shown beneath shared images. This is intended to make the permission model feel like an extension of ordinary messaging behaviour rather than a separate or unfamiliar privacy-management environment. Consistency is therefore treated not only as a visual quality, but also as a way of reducing confusion when users encounter revocation, forwarding, view-once, time-limit, or co-subject controls for the first time.

To achieve NFR8, the prototype is specified as operating alongside familiar WhatsApp-style secure messaging assumptions rather than replacing them. The design does not propose any change to the underlying end-to-end encryption model and does not attempt to solve revocation through cryptographic redesign. Instead, permission management is framed as an additional interface and state-management layer that governs how photos remain visible, reviewable, forwardable, or restricted within the platform context. This preserves conceptual continuity

with users' expectations of secure WhatsApp-like communication while keeping the dissertation focused on interaction and consent management rather than on changes to the underlying encryption architecture.

To achieve NFR9, the prototype communicates the limits of revocation explicitly and avoids implying complete erasure after disclosure. In the current interface, this is expressed through pre-send warning text in the composer, explanatory notes beneath shared photos, revoked-state placeholders, and co-subject request flows that state that screenshots, saved copies, offline retention, and prior viewing cannot be fully undone. The permissions panel also reinforces this boundary by stating that the design is intentionally scoped as platform-level visibility control rather than a guarantee of universal deletion. In this way, the prototype remains honest about the partial and prospective nature of revocation and aims to calibrate user expectations about what its controls can and cannot enforce after a photo has been shared.

Chapter 5

Implementation

5.1 Section Heading

Chapter 6

Legal, Social, Ethical and Professional Issues

Your report should include a chapter with a reasoned discussion about legal, social ethical and professional issues within the context of your project problem. You should also demonstrate that you are aware of the regulations governing your project area and the Code of Conduct & Code of Good Practice issued by the British Computer Society, and that you have applied their principles, where appropriate, as you carried out your project.

6.1 Section Heading

Chapter 7

Results/Evaluation

7.1 Software Testing

7.2 Section Heading

Chapter 8

Conclusion and Future Work

The project's conclusions should list the key things that have been learnt as a consequence of engaging in your project work. For example, "The use of overloading in C++ provides a very elegant mechanism for transparent parallelisation of sequential programs", or "The overheads of linear-time n-body algorithms makes them computationally less efficient than $O(n \log n)$ algorithms for systems with less than 100000 particles". Avoid tedious personal reflections like "I learned a lot about C++ programming...", or "Simulating colliding galaxies can be real fun...". It is common to finish the report by listing ways in which the project can be taken further. This might, for example, be a plan for turning a piece of software or hardware into a marketable product, or a set of ideas for possibly turning your project into an MPhil or PhD.

References

[1] Ioannis Agrafiotis. Enhancing user's privacy : developing a model for managing and testing the lifecycle of consent and revocation. Unpublished, 2012.

[2] Abdulmohsen S. Albesher and Thamer Alhussain. Evaluating and Comparing the Usability of Privacy in WhatsApp, Twitter, and Snapchat. *International Journal of Advanced Computer Science and Applications*, 12(8), 2021.

[3] Lemi Baruh and Mihaela Popescu. Big data analytics and the limits of privacy self-management. *New Media & Society*, 19(4):579–596, April 2017.

[4] Carlos Bermejo Fernandez, Dimitris Chatzopoulos, Dimitrios Papadopoulos, and Pan Hui. This Website Uses Nudging: MTurk Workers' Behaviour on Cookie Consent Notices. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW2):1–22, October 2021.

[5] August Bourgeus and Laurens Vandercruysse. Reinvigorating Consent: Exploring New Paradigms for Privacy and Data Sharing. 2024.

[6] Gary Burkhardt, Frederic Boy, Daniele Doneddu, and Nick Hajli. Privacy Behaviour: A Model for Online Informed Consent. *Journal of Business Ethics*, 186(1):237–255, August 2023.

[7] Vedant Raj Chaurasiya. CONSENT MECHANISMS UNDER THE DIGITAL PERSONAL DATA PROTECTION ACT, 2023: A COMPARATIVE LEGAL ANALYSIS WITH GDPR AND CCPA/CPRA. *LawFoyer International Journal of Doctrinal Legal Research*, 3(2):517–572, June 2025.

[8] Mauro Cherubini, Kavous Salehzadeh Niksirat, Marc-Olivier Boldi, Henri Keopraseuth, Jose M. Such, and Kévin Huguenin. When Forcing Collaboration is the Most Sensible Choice: Desirability of Precautionary and Dissuasive Mechanisms to Manage Mul-

tiparty Privacy Conflicts. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1):1–36, April 2021.

[9] Jeffrey T. Child and Sandra Petronio. Unpacking the paradoxes of privacy in cmc relationships: The challenges of blogging and relational communication on the internet. In Kevin B. Wright and Lynne M. Webb, editors, *Computer-Mediated Communication in Personal Relationships*, chapter 2, pages 21–39. Peter Lang US, January 2011.

[10] Richard L Daft and Robert H Lengel. ORGANIZATIONAL INFORMATION REQUIREMENTS, MEDIA RICHNESS AND STRUCTUR.

[11] Alan R. Dennis and Susan T. Kinney. Testing Media Richness Theory in the New Media: The Effects of Cues, Feedback, and Task Equivocality. *Information Systems Research*, 9(3):256–274, September 1998.

[12] European Union. Regulation (eu) 2016/679 of the european parliament and of the council of 27 april 2016 (general data protection regulation). Official Journal of the European Union, 2016. OJ L 119, 4 May 2016, pp. 1–88.

[13] Colin M. Gray, Cristiana Santos, Nataliia Bielova, Michael Toth, and Damian Clifford. Dark Patterns and the Legal Requirements of Consent Banners: An Interaction Criticism Perspective. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pages 1–18, May 2021. arXiv:2009.10194 [cs].

[14] Paul Graßl, Hanna Schraffenberger, Frederik Zuiderveen Borgesius, and Moniek Buijzen. Dark and bright patterns in cookie consent requests. July 2020.

[15] Howard Heath, Áine MacDermott, and Alex Akinbi. Forensic analysis of ephemeral messaging applications: Disappearing messages or evidential data? *Forensic Science International: Digital Investigation*, 46:301585, September 2023.

[16] Susan C. Herring. The coevolution of computer-mediated communication and computer-mediated discourse analysis. In Patricia Bou-Franch and Pilar Garcés-Conejos Blitvich, editors, *Analyzing Digital Discourse*, pages 25–67. Springer International Publishing, Cham, 2019.

[17] Jane Im, Jill Dimond, Melody Berton, Una Lee, Katherine Mustelier, Mark S. Ackerman, and Eric Gilbert. Yes: Affirmative Consent as a Theoretical Framework for Understanding

and Imagining Social Platforms. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pages 1–18, Yokohama Japan, May 2021. ACM.

[18] Alexis Shore Ingber. Understanding screenshot collection and sharing on messaging platforms: a privacy perspective. *Journal of Computer-Mediated Communication*, 30(1):zmae023, January 2025.

[19] Kumi Ishii, Mary Madison Lyons, and Sabrina A. Carr. Revisiting media richness theory for today and future. *Human Behavior and Emerging Technologies*, 1(2):124–131, April 2019.

[20] Mizuko Ito. Mobile Phones, Japanese Youth, and the Re-placement of Social Contact. In *Mobile Communications*, pages 131–148. Springer-Verlag, London, 2005. Series Title: Computer Supported Cooperative Work.

[21] Gayatri Priyadarsini Kancherla, Nataliia Bielova, Cristiana Santos, and Abhishek Bichhawat. Measuring Compliance of Consent Revocation on the Web, May 2025. arXiv:2411.15414 [cs].

[22] Jane Kaye, Edgar A Whitley, David Lund, Michael Morrison, Harriet Teare, and Karen Melham. Dynamic consent: a patient interface for twenty-first century research networks. *European Journal of Human Genetics*, 23(2):141–146, February 2015.

[23] Tim Kindberg, Mirjana Spasojevic, Rowanne Fleck, and Abigail Sellen. I saw this and thought of you: some social uses of camera phones. In *CHI '05 Extended Abstracts on Human Factors in Computing Systems*, pages 1545–1548, Portland OR USA, April 2005. ACM.

[24] Ah Ra Lee, Dongjun Koo, Il Kon Kim, Eunjoo Lee, Hyun Ho Kim, Sooyoung Yoo, Jeong-Hyun Kim, Eun Kyung Choi, and Ho-Young Lee. Identifying facilitators of and barriers to the adoption of dynamic consent in digital health ecosystems: a scoping review. *BMC Medical Ethics*, 24(1):107, December 2023.

[25] Daniel Lindegren, Farzaneh Karegar, Bridget Kane, and John Sören Pettersson. An evaluation of three designs to engage users when providing their consent on smartphones. *Behaviour & Information Technology*, 40(4):398–414, March 2021.

[26] Chi Liu, Tianqing Zhu, Jun Zhang, and Wanlei Zhou. Privacy Intelligence: A Survey on Image Privacy in Online Social Networks. *ACM Computing Surveys*, 55(8):1–35, August 2023. arXiv:2008.12199 [cs].

[27] Mpyana Mwamba Merlec, Youn Kyu Lee, Seng-Phil Hong, and Hoh Peter In. A Smart Contract-Based Dynamic Consent Management System for Personal Data Usage under GDPR. *Sensors*, 21(23):7994, November 2021.

[28] Minal R. Narkhede, Nilesh I. Wankhede, and Akanksha M. Kamble. Enhancing patient autonomy in data ownership: privacy models and consent frameworks for healthcare. *Journal of Digital Health*, pages 1–23, March 2025.

[29] Helen Nissenbaum. A Contextual Approach to Privacy Online. 2011.

[30] Midas Nouwens, Ilaria Liccardi, Michael Veale, David Karger, and Lalana Kagal. Dark Patterns after the GDPR: Scraping Consent Pop-ups and Demonstrating their Influence. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, pages 1–13, April 2020. arXiv:2001.02479 [cs].

[31] Sandra Petronio and Jeffrey T Child. Conceptualization and operationalization: utility of communication privacy management theory. *Current Opinion in Psychology*, 31:76–82, February 2020.

[32] Eugenia Politou, Efthimios Alepis, and Constantinos Patsakis. Forgetting personal data and revoking consent under the GDPR: Challenges and proposed solutions. *Journal of Cybersecurity*, 4(1), January 2018.

[33] Marco Robol, Travis D. Breaux, Elda Paja, and Paolo Giorgini. Consent Verification Monitoring. *ACM Transactions on Software Engineering and Methodology*, 32(1):1–33, January 2023.

[34] Florian Schaub, Rebecca Balebako, Adam Durity, and Lorrie Cranor. A design space for effective privacy notices. July 2015.

[35] Theodor Schnitzler, Christine Utz, Florian M Farke, Christina Pöpper, and Markus Dürmuth. Exploring user perceptions of deletion in mobile instant messaging applications. *Journal of Cybersecurity*, 6(1):tyz016, January 2020.

[36] Arianna Schuler Scott. *Dynamic Consent: a Mechanism for Engagement*. PhD thesis, University of Oxford, February 2022.

[37] John Short, Ederyn Williams, and Bruce Christie. *The Social Psychology of Telecommunications*. John Wiley & Sons, London, 1976.

[38] Than Htut Soe, Oda Elise Nordberg, Frode Guribye, and Marija Slavkovik. Circumvention by design – dark patterns in cookie consents for online news outlets, June 2020. arXiv:2006.13985 [cs].

[39] Ari Ezra Waldman. Designing Without Privacy.

[40] WhatsApp. About advanced chat privacy. https://faq.whatsapp.com/715385484388016. WhatsApp Help Center. Retrieved December 16, 2025.

[41] WhatsApp. About disappearing messages. https://faq.whatsapp.com/673193694148537. WhatsApp Help Center. Retrieved December 16, 2025.

[42] WhatsApp. About end-to-end encryption. https://faq.whatsapp.com/820124435853543. WhatsApp Help Center. Retrieved December 16, 2025.

[43] WhatsApp. About view once. https://faq.whatsapp.com/1077018839582332. WhatsApp Help Center. Retrieved December 16, 2025.

[44] WhatsApp. How to delete messages. https://faq.whatsapp.com/1370476507114859. WhatsApp Help Center. Retrieved December 16, 2025.

[45] WhatsApp. Introducing advanced chat privacy: Enhanced protection for your most sensitive conversations. https://blog.whatsapp.com/introducing-advanced-chat-privacy?lang=en, April 2025. WhatsApp Blog.

[46] WhatsApp LLC. Privacy policy – uk. https://www.whatsapp.com/legal/privacy-policy-uk, March 2025.

[47] John Wilbanks. Design Issues in E-Consent. *Journal of Law, Medicine & Ethics*, 46(1):110–118, 2018.

[48] Kaihe Xu, Yuanxiong Guo, Linke Guo, Yuguang Fang, and Xiaolin Li. My Privacy My Decision: Control of Photo Sharing on Online Social Networks. *IEEE Transactions on Dependable and Secure Computing*, 14(2):199–210, March 2017.

[49] Lei Xu, Ting Bao, Liehuang Zhu, and Yan Zhang. Trust-Based Privacy-Preserving Photo Sharing in Online Social Networks. *IEEE Transactions on Multimedia*, 21(3):591–602, March 2019.

[50] Ming Zhang, Zhe Sun, Hui Li, Ben Niu, Fenghua Li, Zixu Zhang, Yuhang Xie, and Chunhao Zheng. Go-Sharing: A Blockchain-Based Privacy-Preserving Framework for Cross-Social Network Photo Sharing. *IEEE Transactions on Dependable and Secure Computing*, 20(5):3572–3587, September 2023.

Appendix A

Extra Information

A.1 Tables, proofs, graphs, test cases, ...

The appendices contain information that is peripheral to the main body of the report. Information typically included in the Appendix are things like tables, proofs, graphs, test cases or any other material that would break up the theme of the text if it appeared in the body of the report. It is necessary to include your source code listings in an appendix that is separate from the body of your written report (see the information on Program Listings below).

A.2 Survey Information Sheet

A.3 Survey Questions

Appendix B

User Guide

B.1 Instructions

You must provide an adequate user guide for your software. The guide should provide easily understood instructions on how to use your software. A particularly useful approach is to treat the user guide as a walk-through of a typical session, or set of sessions, which collectively display all of the features of your package. Technical details of how the package works are rarely required. Keep the guide concise and simple. The extensive use of diagrams, illustrating the package in action, can often be particularly helpful. The user guide is sometimes included as a chapter in the main body of the report, but is often better included in an appendix to the main report.

Appendix C

Source Code

C.1 Instructions

Complete source code listings must be submitted as an appendix to the report. The project source codes are usually spread out over several files/units. You should try to help the reader to navigate through your source code by providing a "table of contents" (titles of these files/units and one line descriptions). The first page of the program listings folder must contain the following statement certifying the work as your own: "I verify that I am the sole author of the programs contained in this folder, except where explicitly stated to the contrary". Your (typed) signature and the date should follow this statement.

All work on programs must stop once the code is submitted to KEATS. You are required to keep safely several copies of this version of the program and you must use one of these copies in the project examination. Your examiners may ask to see the last-modified dates of your program files, and may ask you to demonstrate that the program files you use in the project examination are identical to the program files you have uploaded to KEATS. Any attempt to demonstrate code that is not included in your submitted source listings is an attempt to cheat; any such attempt will be reported to the KCL Misconduct Committee.

You may find it easier to firstly generate a PDF of your source code using a text editor and then merge it to the end of your report. There are many free tools available that allow you to merge PDF files.